

Stefan Bobrowski

Full-Stack AI Engineer

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Professional Summary

Full-Stack AI Engineer with over 10 years of experience building intelligent, scalable web applications that integrate cutting-edge AI capabilities. Expert in React, TypeScript, and modern JavaScript development, with hands-on experience implementing AI-powered features using Claude, Gemini, and Vertex AI. Skilled in architecting secure, cloud-native solutions on Google Cloud Platform, developing both elegant user interfaces with Three.js/WebGL and robust backend APIs with Node.js/Express. Passionate about leveraging AI to solve real-world problems, writing clean and maintainable code, and delivering innovative solutions that provide measurable business value.

Technical Skills

Frontend

JavaScript, TypeScript, React, Vite, Astro, Zustand, TanStack Query (React Query), Axios, Lottie, SASS/SCSS, CSS Modules, Responsive Design, WCAG Accessibility

Backend

Node.js, Express, REST APIs, GraphQL, WebSockets, PostgreSQL, Passport.js, Multer, Postman

AI Tools

GitHub Copilot, Claude, Gemini, Vertex AI, ChatGPT

Cloud

Google Cloud Platform, Cloud Run, Cloud Storage, Cloud SQL, Artifact Registry, BigQuery

DevOps

Git, GitHub, GitHub Actions, Cloud Build, Docker, Kubernetes

Security

IAM, WIF (Workload Identity Federation), Rate Limiting, reCAPTCHA v3, OAuth 2.0, JWT, Helmet, CORS

Visualization

Three.js, React Three Fiber, Drei, D3.js, Chart.js, HTML5 Canvas, WebGL

Design

Figma, Adobe Photoshop, Adobe Premiere Pro

Game Dev

Unreal Engine 5

Professional Experience

Frontend Developer, Camelot SI LLC, SharperImage.com

Sep 2023 - Feb 2025

- Spearheaded the redesign of key CRO components—including checkout, cart, login, VIP account, returns portal, and site-wide layout (header/footer)—within a modern Next.js e-commerce architecture.
- Contributed to Bitbucket Pipelines CI/CD workflow across dev, staging, and production; regularly merged pull requests, cut releases, and triggered early-morning deployments and hotfixes.

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- Contributed to Bitbucket Pipelines CI/CD workflow across dev, staging, and production; regularly merged pull requests, cut releases, and triggered early-morning deployments and hotfixes.
- Reduced frontend bundle size from ~50MB to ~12MB by auditing and removing unused packages, upgrading dependencies, and optimizing build configs—dramatically improving pipeline reliability and build speed.
- Consolidated GraphQL requests into a single batched query, improving frontend performance and reducing API overhead.
- Integrated Contentful-based feature flags and A/B tests for dynamic marketing campaigns.
- Delivered features and bug fixes for the internal Merchant Center dashboard used by warehouse and customer support teams; implemented a Git Flow branching strategy and documented deployment process in Confluence.

Software Engineer, The Home Depot

May 2021 - Dec 2022

- Led development of "Questions & Answers" component redesign (Figma, React, GraphQL), improving engagement and CTR.
- Served as Release Captain; reviewed PRs, wrote release notes, and deployed to production.
- Increased test coverage to 100% for several components by transitioning from Enzyme to RTL + Jest.
- Onboarded and mentored new engineers on best practices within a large organization.

Software Developer, Vibrant Software

July 2020 - May 2021

- Led development of the Nomad Portal, a content management system using React + Redux-Saga and powered by AWS AI/ML services. Featured dynamically generated rich asset metadata and tags, advanced search with filters, and content facial recognition.
- Built VEP Virtual Stage a live streaming platform that has been used successfully for live virtual events and concerts. Featured multiple stream support, live chat, donation form, and fully configurable content from JSON.

Full-Stack Web Developer, Hat Trick Designs LLC

June 2019 - July 2020

- Developed full-stack apps with the MERN stack.
- Frontend React with Hooks and state management with useContext, backend REST API with Express and Mongoose.

Front-end Web Developer, Direct Wines

February 2019 - May 2019

- Developed product microsites and customer checkout component for creating, retrieving, updating, and deleting payment methods (JavaScript, Fetch, async/await).
- Improved SEO and accessibility with metadata and Structured Data markup, allowing for search results to display custom widgets such as reviews and featured links.
- Researched and tested emerging front end technologies such as Web Components, Vue, and React with Hooks for use in future code base.

Front-end Web Developer, Kathy Kuo Home

December 2017 to September 2018

- Prototyped new features and weighed UI/UX decisions with Design Lead and CEO.
- Created sister-site (<https://www.kathykuohome.com/designbar>), successfully displaying business portfolio to clients.
- Implemented Google Analytics and Google Optimize A/B tests, resulting in an increased click-through rate for new UI design changes.
- Increased Google PageSpeed Insights score by 300% and reduced average page load time by 3 seconds.

Front-end Web Developer, Archi FX LLC, Healthcasts, Finalsight

September 2014 - December 2017

- Archi FX LLC: Led front end development on new responsive websites (Flexbox).
- Healthcasts: Implemented product microsites containing sensitive medical diagnostic information (Node.js, NPM scripts, jQuery, animations).
- Finalsight: Succeeded in a fast-paced Agile environment building many responsive websites for schools globally (SASS, jQuery, custom CMS).

Web Developer Freelancer, oDesk

November 2007 - July 2014

- Worked on various freelance projects primarily with HTML, CSS, and JavaScript.

Education

Bachelor of Science in Computer Science, University of Connecticut

September 2010 - May 2014